

Project Report

on

Web Security & Threat

Intelligence

Platform(SecureGaurd)

By

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ACKNOWLEDGEMENT

We take great pleasure in presenting our project titled “Web Security Platform (SecureGuard)”, completed as part of the academic curriculum for the degree of Master of Computer Applications.

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We are also thankful to all the faculty members of the department for providing us with the necessary knowledge and resources required for the completion of this project.

We extend our heartfelt thanks to our friends and classmates for their support, cooperation, and valuable suggestions during the project development phase.

Finally, we would like to thank our family members for their constant motivation and encouragement, which helped us complete this project successfully.

This project has helped us gain practical knowledge in web technologies, database management, and cybersecurity concepts, and has been a valuable learning experience for us.

Project Team

Web Security Platform (SecureGuard)

Master of Computer Applications

Academic Year: 2025–26

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CHAPTER 1: INTRODUCTION

1.1 Existing System and Need for System

In the existing system, users manually check suspicious URLs and messages without proper tools. There is no centralized platform to verify whether a link or message is safe or malicious.

This leads to:

- High risk of phishing and cyber attacks
- Lack of awareness about online threats
- No proper tracking of suspicious activities
- Time-consuming manual verification

Also, most users do not have technical knowledge to identify cyber threats effectively.

Hence, there is a need for a **web-based intelligent system** that can:

- Detect malicious URLs and messages
 - Provide risk classification
 - Maintain scan history
 - Help users stay safe online
-

1.2 Scope of Work

The project includes the development of a web-based platform for detecting cyber threats.

Work is divided into modules:

- User Interface Module
- Security Analysis Module
- Data Management Module
- Admin Control Module

My responsibilities include:

- Designing system logic
 - Developing scanning modules
 - Managing database operations
 - Implementing user authentication
-

1.3 Operating Environment

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Hard Disk: 100 GB

Software Requirements:

- Operating System: Windows 10 / 11
 - Framework: ASP.NET Core (.NET 6/7)
 - Web Server: IIS
 - Database: SQL Server
 - Browser: Chrome / Edge
-

CHAPTER 2: PROPOSED SYSTEM

2.1 Proposed System

The proposed system is a *Web Security & Threat Intelligence Platform* that helps users detect and prevent cyber threats.

Modules:

- **User Registration Module**
Allows users to create and manage accounts
 - **URL Scanning Module**
Checks links and classifies them as Safe / Suspicious / Dangerous
 - **Message Scanning Module**
Detects spam and phishing messages
 - **Scan History Module**
Stores previous scans for users
 - **Report Module**
Allows reporting malicious content
 - **Admin Module**
Manages users, threats and system data
-

2.2 Objectives of System

- Reduce cyber security risks
 - Provide quick and accurate threat detection
 - Improve user awareness
 - Maintain centralized threat database
 - Provide user-friendly interface
 - Enable real-time scanning
-

2.3 User Requirements

User:

- Register and login
- Scan URLs and messages
- View scan results
- Access scan history
- Report threats

Admin:

- Manage users
- Monitor system
- Review reports
- Block malicious content

Guest:

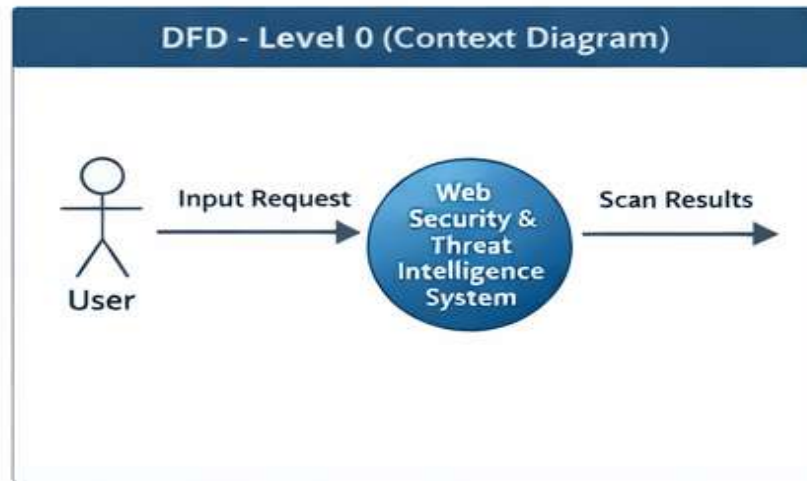
- Scan URLs/messages
 - Limited access
-

CHAPTER 3: ANALYSIS & DESIGN

3.1 DFD

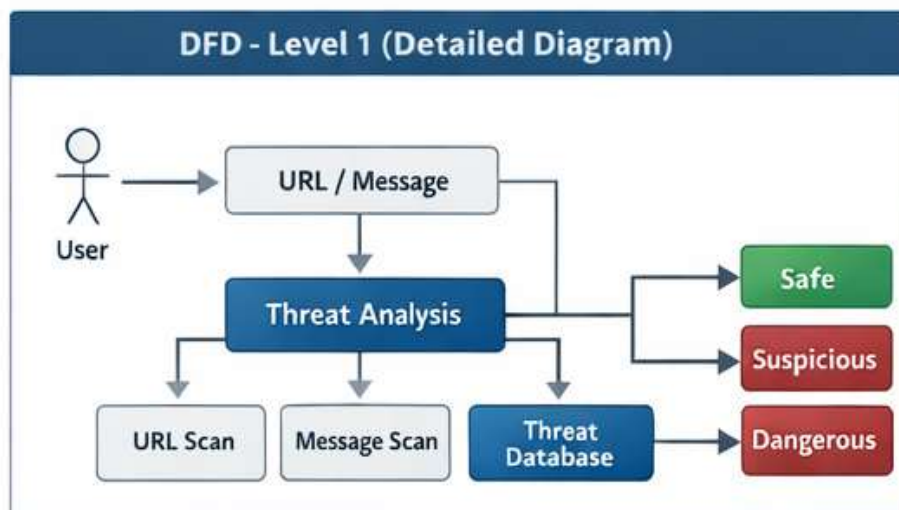
- **Level 0 (Context Diagram):**

User interacts with system → system processes → returns result



- **Level 1:**

- Input: URL / Message
- Process: Threat Analysis
- Output: Risk classification



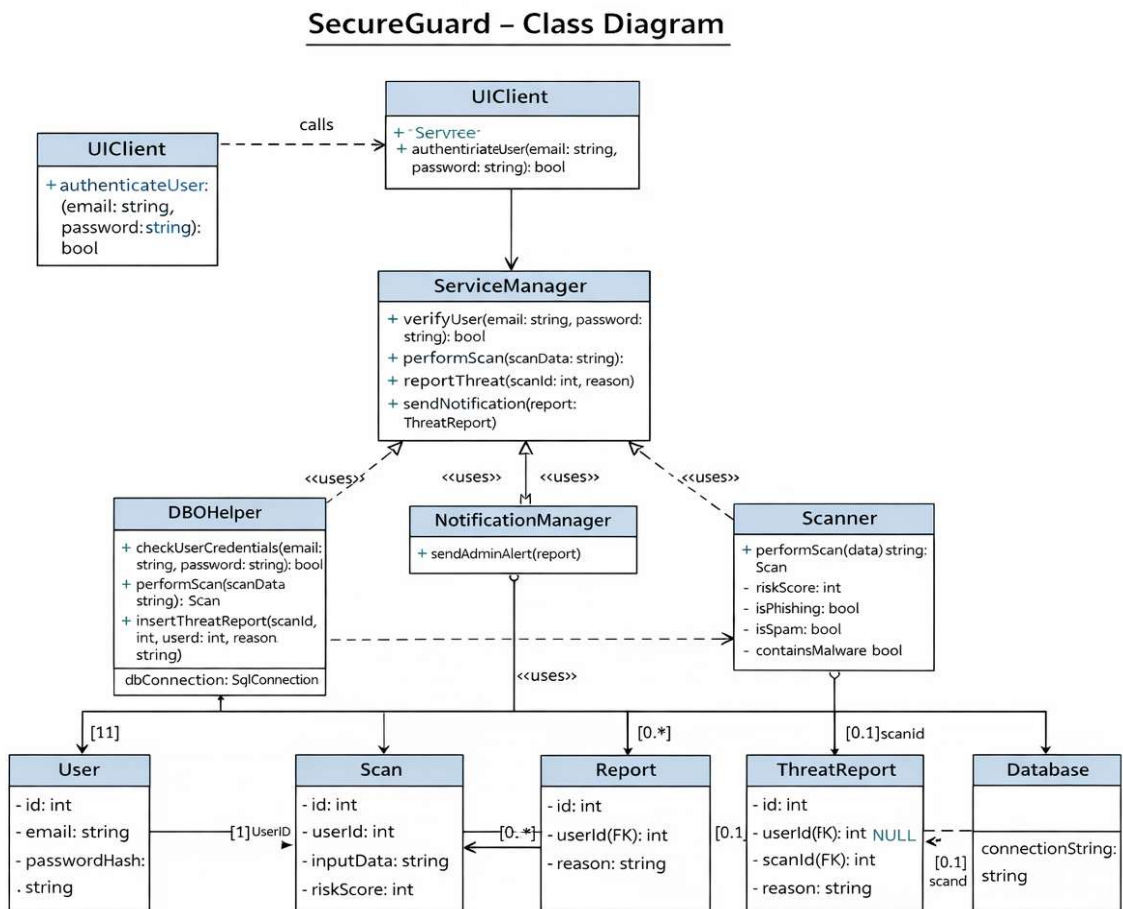
3.2 Functional Decomposition Diagram

Main functions:

- Authentication
- URL Analysis
- Message Analysis
- Reporting System
- Admin Control

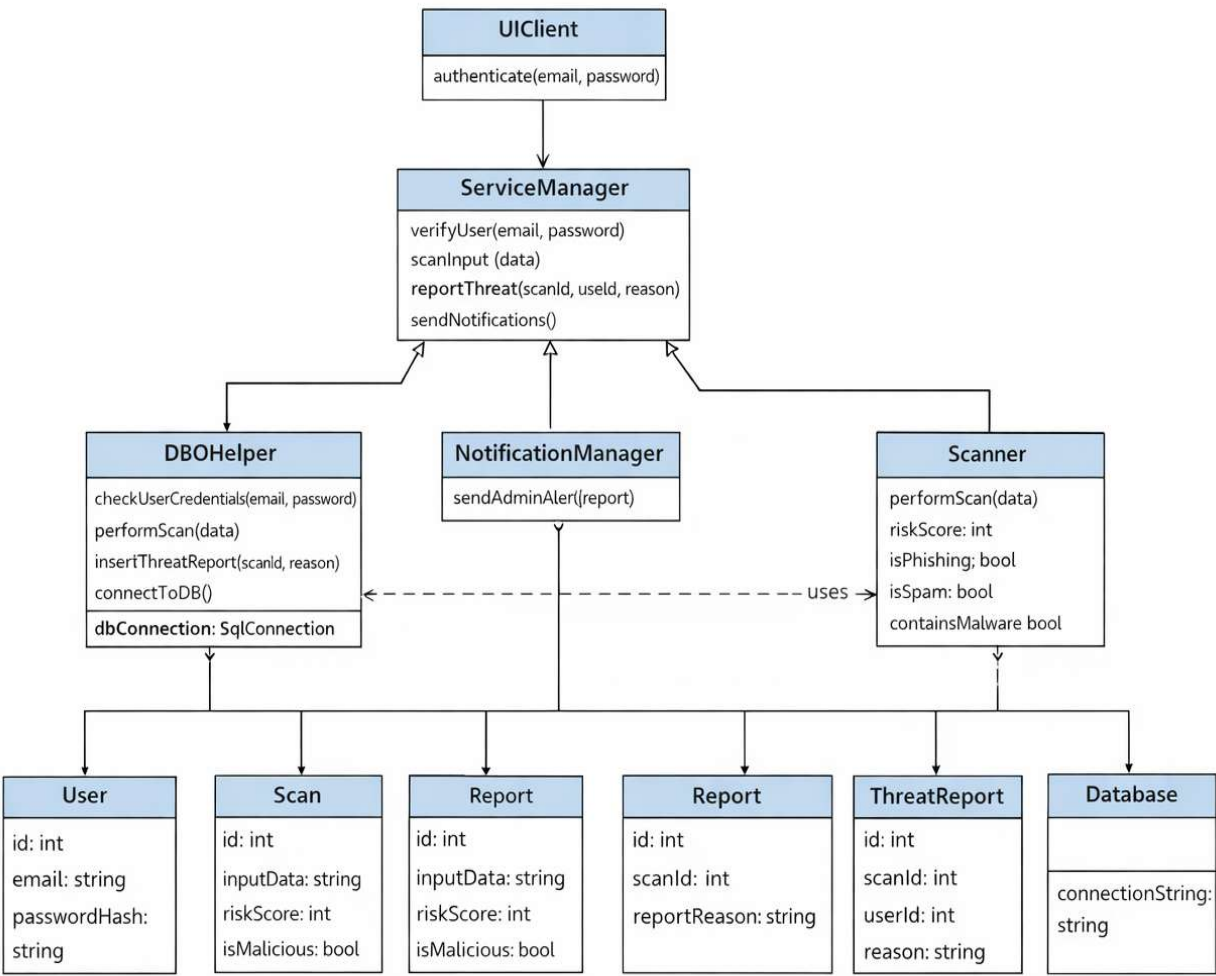
3.3 UML

3.3.1 Class Diagram

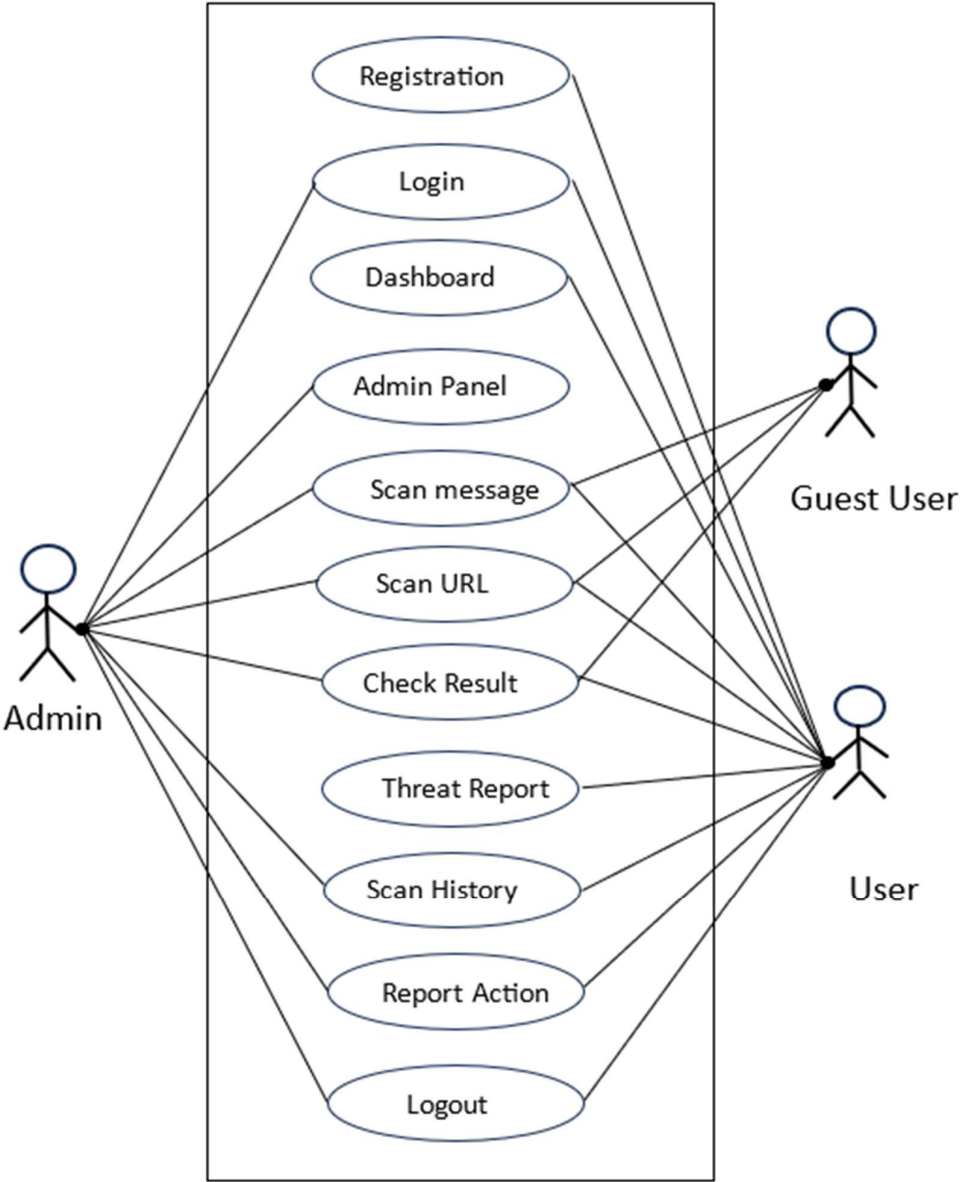


3.3.2 Object Diagram

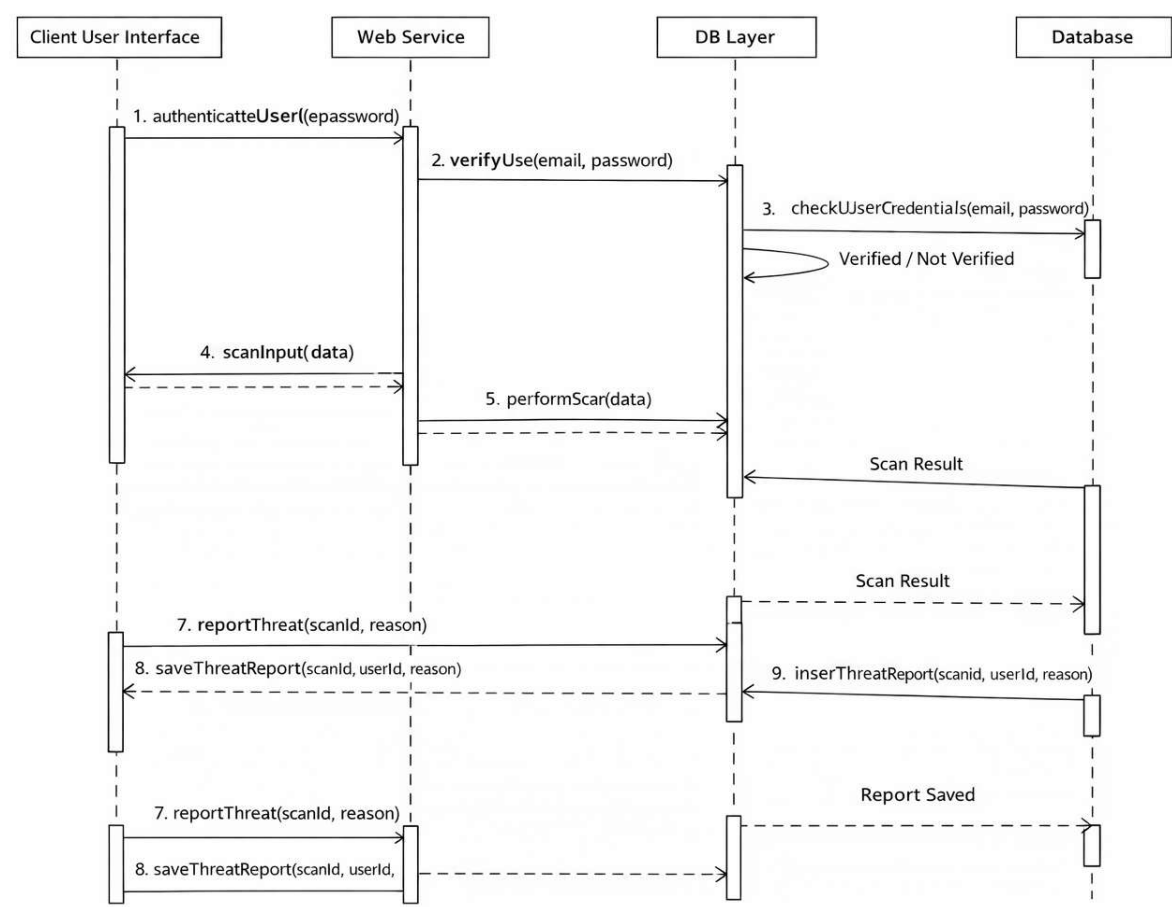
SecureGuard – Object Diagram



3.3.3 Use Case Diagram

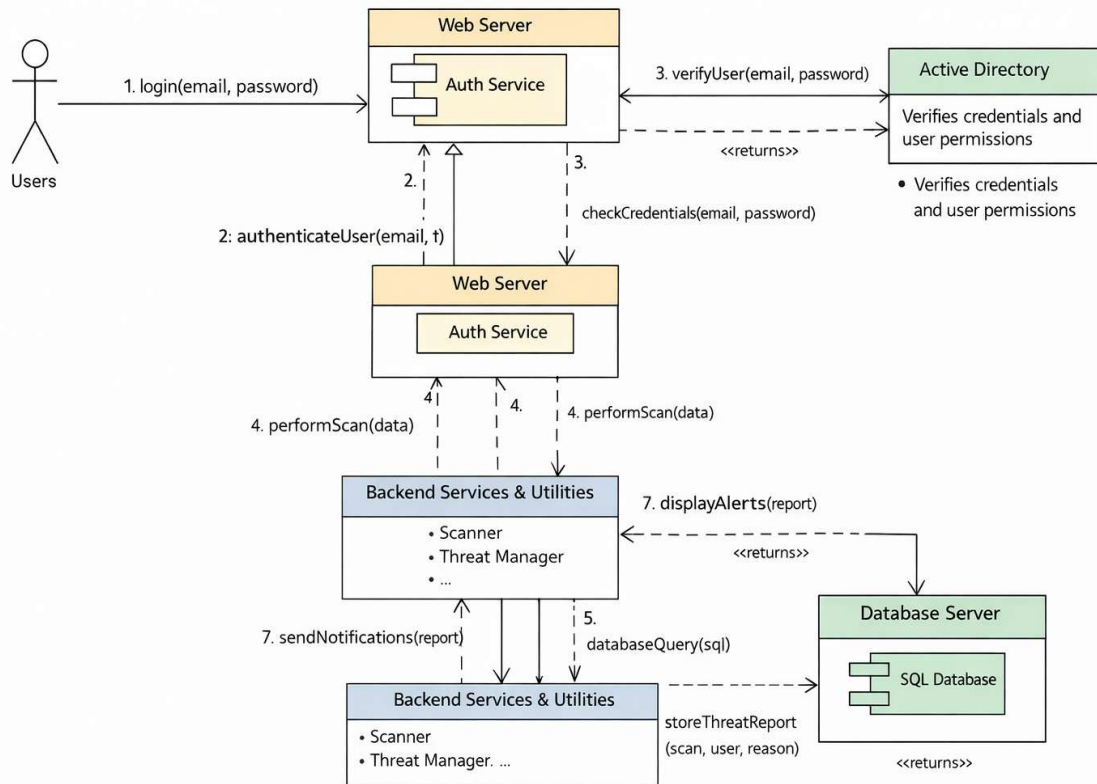


3.3.4 Sequence Diagram

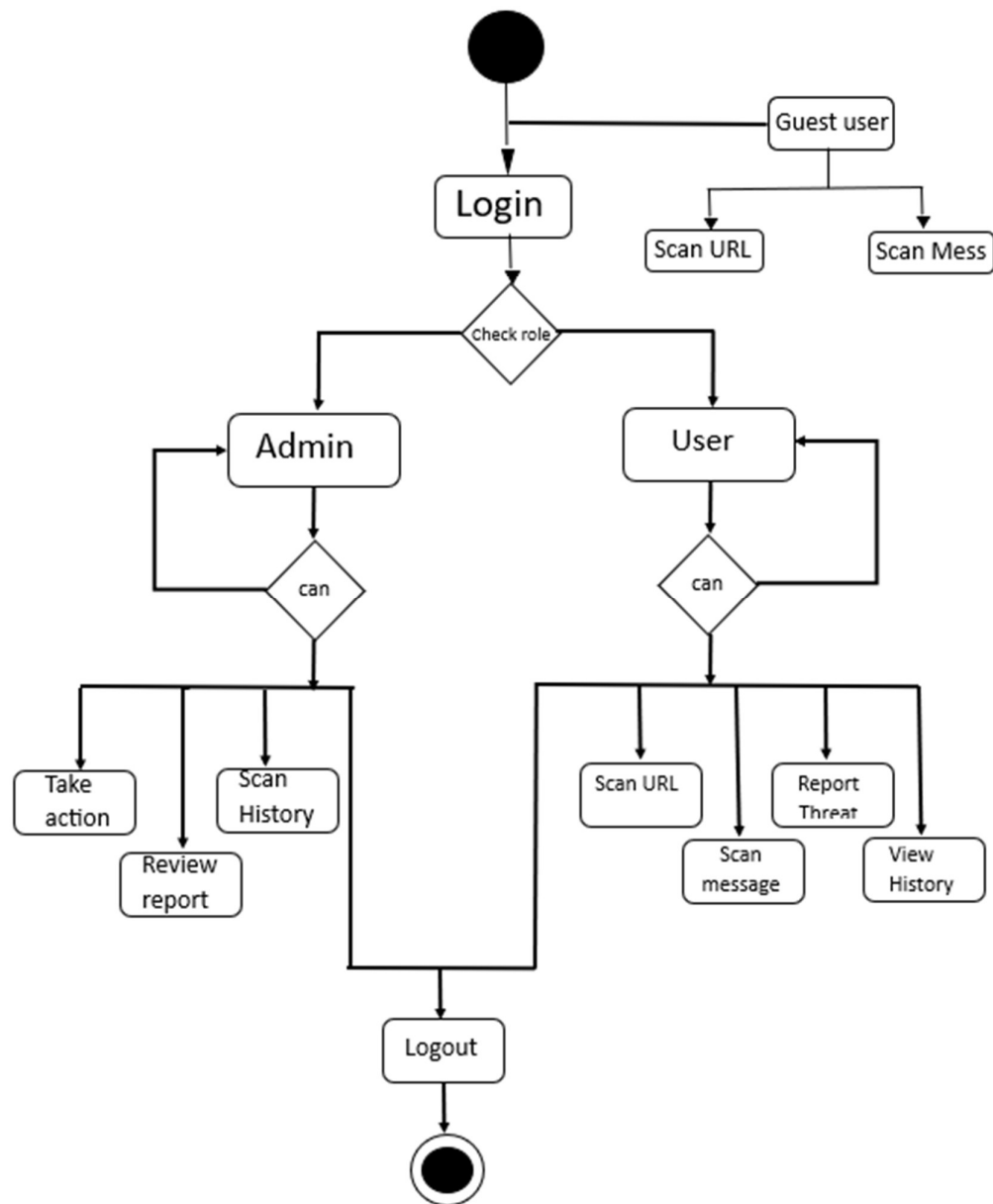


3.3.5 Collaboration Diagram

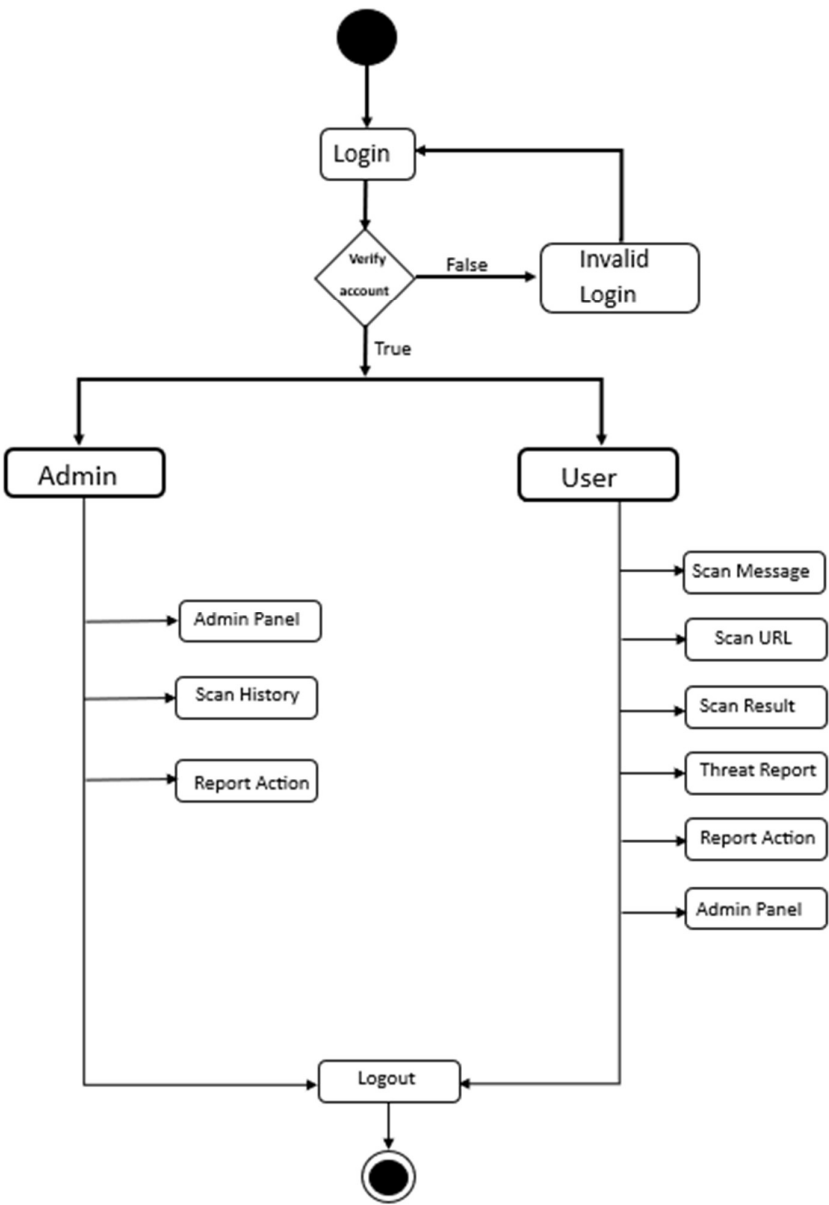
SecureGuard – Collaboration Diagram



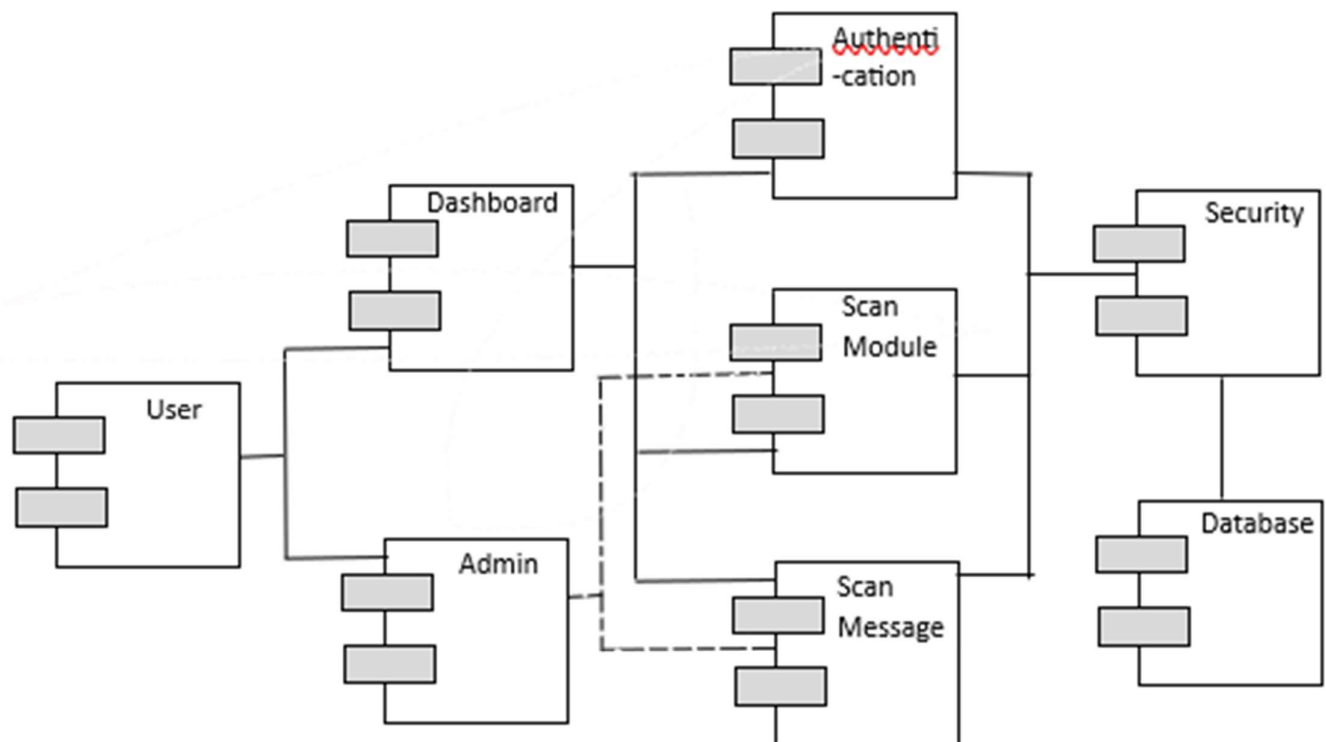
3.3.6 State Diagram



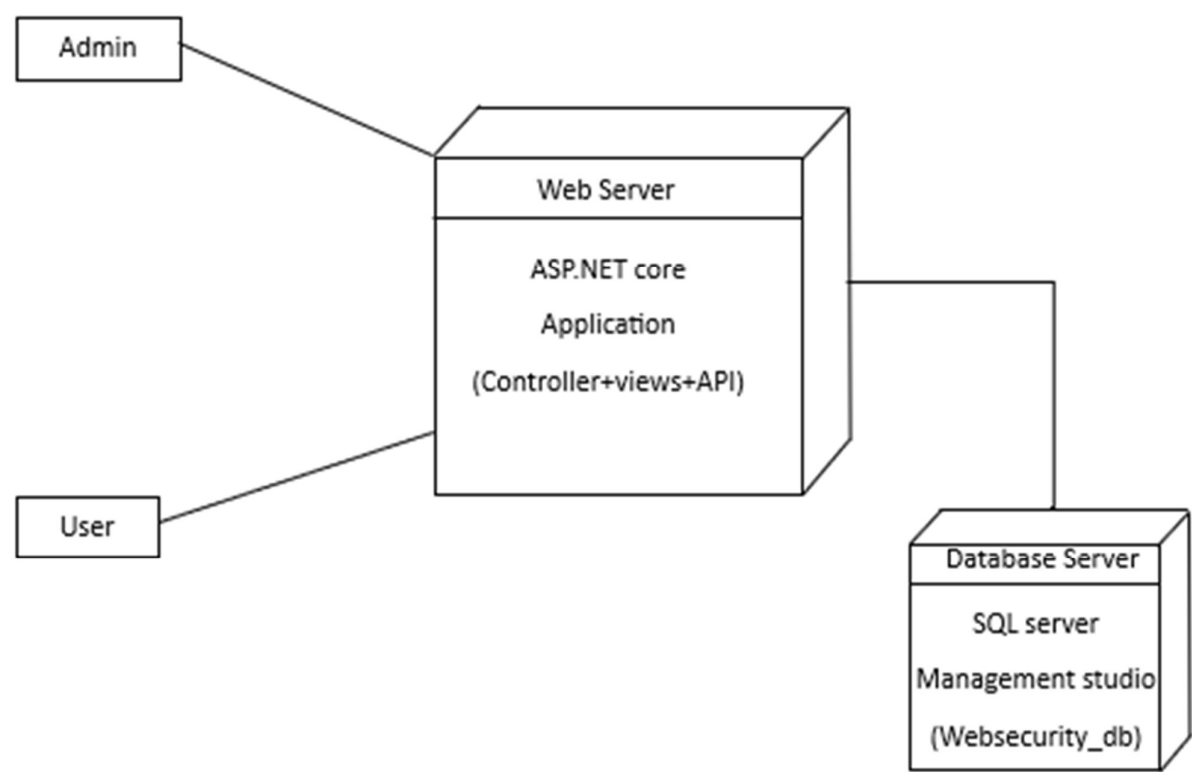
3.3.7 Activity Diagram



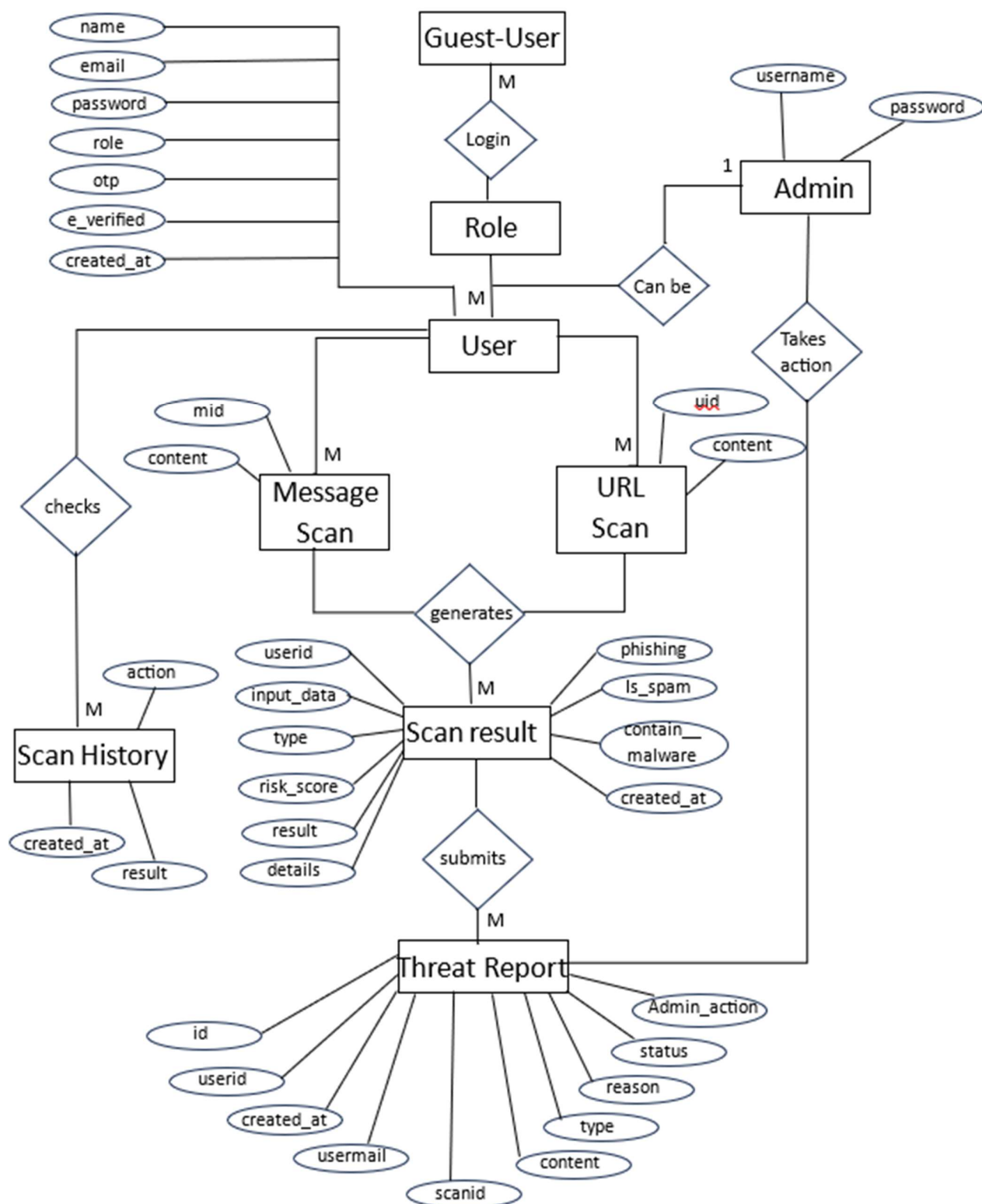
3.3.8 Component Diagram



3.3.9 Deployment Diagram



3.4 ER Diagram



3.5 Data Dictionary

Table	Description
Users	Stores user details
Scans	Stores scan results
Reports	Stores reported threats
Admin	Admin credentials

Table: Users

Column Name	Data Type	Constraints	Description
Id	int	Primary Key	Unique user ID
FullName	varchar(100)	Not Null	User full name
Email	varchar(100)	Unique, Not Null	User email address
PasswordHash	varchar(255)	Not Null	Encrypted password
Role	varchar(20)	Default: User	User role (User/Admin)
OTP	varchar(10)	Nullable	One Time Password
OTPExpiry	datetime	Nullable	OTP expiry time
IsEmailVerified	bit	Default: False	Email verification status
PasswordResetToken	varchar(255)	Nullable	Token for password reset
TokenExpiry	datetime	Nullable	Token expiry time
CreatedAt	datetime		Account creation timestamp

Table: Scans

Column Name	Data Type	Constraints	Description
Id	int	Primary Key	Unique scan ID
UserId	int	Foreign Key	References Users(Id)
InputData	varchar(255)	Not Null	URL or message input
Type	varchar(20)		Scan type (URL/Message)
RiskScore	int		Risk percentage

Result	varchar(20)		Safe / Suspicious / Dangerous
Details	varchar(255)	Nullable	Scan explanation
IsPhishing	bit	Default: False	Phishing detection flag
IsSpam	bit	Default: False	Spam detection flag
ContainsMalware	bit	Default: False	Malware detection flag
CreatedAt	datetime		Scan timestamp

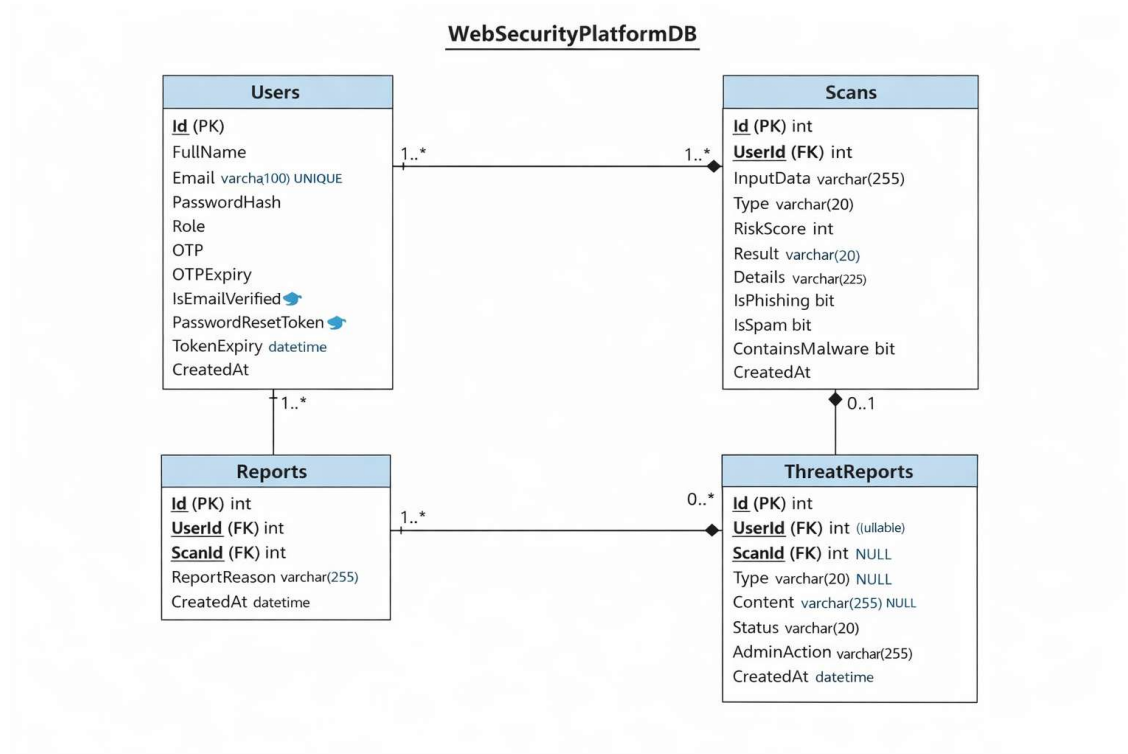
Table: Reports

Column Name	Data Type	Constraints	Description
Id	int	Primary Key	Unique report ID
UserId	int	Foreign Key	References Users(Id)
ScanId	int	Foreign Key	References Scans(Id)
ReportReason	varchar(255)	Not Null	Reason for reporting
CreatedAt	datetime		Report submission date

Table: ThreatReports

Column Name	Data Type	Constraints	Description
Id	int	Primary Key	Unique threat report ID
UserId	int	Nullable	References Users(Id)
UserEmail	varchar(100)	Nullable	Email of reporter
ScanId	int	Nullable	Related scan ID
Type	varchar(20)	Nullable	Type (URL/Message)
Content	varchar(255)	Nullable	Suspicious content
Reason	varchar(255)	Nullable	Reason for report
Status	varchar(20)	Default: Pending	Report status
AdminAction	varchar(255)	Nullable	Action taken by admin
CreatedAt	datetime		Report creation date

3.6 Table Design



3.7 Code Design

```
1 namespace WebSecurityPlatform.Models
2 {
3     public class Scan
4     {
5         public int Id { get; set; }
6         public int UserId { get; set; }
7         public User? User { get; set; }
8
9         public string InputData { get; set; }
10        public string Type { get; set; }
11
12        public int RiskScore { get; set; }
13        public string Result { get; set; }
14        public string? Details { get; set; }
15
16        public bool IsPhishing { get; set; } = false;
17        public bool IsSpam { get; set; } = false;
18        public bool ContainsMalware { get; set; } = false;
19
20        public DateTime CreatedAt { get; set; } = DateTime.Now;
21    }
22 }
```

EXPLORER

WebSecurityPlatform

WebSecurityPlatform

bin

Controllers

Migrations

Models

ApplicationDbContext.cs

DashboardViewModel.cs

EmailService.cs

ErrorViewModel.cs

ForgotPasswordViewModel.cs

HistoryController.cs

LoginViewModel.cs

RegisterViewModel.cs

Report.cs

Scan.cs

SecurityScanService.cs

ThreatReport.cs

User.cs

obj

Properties

Views

Admin

Index.cshtml

Reports.cshtml

@ Index.cshtml X

WebSecurityPlatform > WebSecurityPlatform > Views > Admin > @ Index.cshtml

```
1 @{
2     ViewData["Title"] = "Admin Panel";
3 }
4
5 <h2 class="text-success">👑 Admin Control Panel</h2>
6 <p>System overview and threat management</p>
7
8 <!-- 📊 STATS -->
9 <div class="row mt-4 text-center">
10
11     <div class="col-md-3">
12         <div class="card bg-dark text-white p-3">
13             <h5>👤 Total Users</h5>
14             <h2>@ViewBag.TotalUsers</h2>
15         </div>
16     </div>
17
18     <div class="col-md-3">
19         <div class="card bg-dark text-white p-3">
20             <h5>🔍 Total Scans</h5>
21             <h2>@ViewBag.TotalScans</h2>
22         </div>
23     </div>
24
25     <div class="col-md-3">
26         <div class="card bg-dark text-white p-3">
27             <h5>🚨 Reported Threats</h5>
28             <h2>@ViewBag.TotalReports</h2>
29         </div>
30     </div>
31 </div>
```

EXPLORER

WebSecurityPlatform

WebSecurityPlatform

.vs

WebSecurityPlatform

bin

Controllers

AdminController.cs

AuthController.cs

HomeController.cs

UserController.cs

Migrations

Models

ApplicationDbContext.cs

DashboardViewModel.cs

EmailService.cs

ErrorViewModel.cs

ForgotPasswordViewModel.cs

HistoryController.cs

LoginViewModel.cs

RegisterViewModel.cs

Report.cs

Scan.cs

SecurityScanService.cs

ThreatReport.cs

User.cs

C# AdminController.cs X

WebSecurityPlatform > WebSecurityPlatform > Controllers > C# AdminController.cs

```
1 using Microsoft.AspNetCore.Mvc;
2 using Microsoft.AspNetCore.Authorization;
3 using Microsoft.EntityFrameworkCore; // 🔥 IMPORTANT
4 using WebSecurityPlatform.Models;
5
6 [Authorize(Roles = "Admin")]
7 public class AdminController : Controller
8 {
9     private readonly ApplicationDbContext _context;
10
11     public AdminController(ApplicationDbContext context)
12     {
13         _context = context;
14     }
15
16     // ===== DASHBOARD =====
17     public IActionResult Index()
18     {
19         // 📊 COUNTS
20         ViewBag.TotalUsers = _context.Users.Count();
21         ViewBag.TotalScans = _context.Scans.Count();
22
23         ViewBag.TotalReports = _context.ThreatReports.Count();
24         ViewBag.ResolvedReports = _context.ThreatReports
25             .Count(r => r.Status == "Resolved");
26
27         // 🚨 REPORTED THREATS (SAFE NULL HANDLING)
28         var reports = _context.ThreatReports
```

CHAPTER 4: USER MANUAL

4.1 Operations

- User registers and logs in
 - Enters URL/message
 - System analyses data
 - Result displayed
 - User can report threats
-

4.2 Menu Screens (At end)

- Login Page
 - Dashboard
 - Scan Page
 - History Page
 - Admin Panel
-

4.3 Project Code

The system is developed using ASP.NET Core with SQL Server backend and follows modular architecture.

CHAPTER 5: Testing

5.1 Testing Test Plan

The testing process ensures that the system works correctly and meets user requirements. The testing strategy includes validating all modules such as URL scanning, message scanning, user authentication, and reporting.

Objectives:

- To verify system functionality
- To detect errors and bugs
- To ensure system reliability and security
- To validate user inputs and outputs

Testing Types Used:

- Black Box Testing
 - White Box Testing
 - Data Validation Testing
-

5.2 Black Box Testing (Data Validation)

Black box testing focuses on testing system functionality without knowing internal code logic.

Key Areas Tested:

- URL input validation
- Message scanning accuracy
- Login authentication
- Form validations
- Output correctness

Example:

Input	Expected Output	Actual Output	Status
Valid URL	Safe / Suspicious / Dangerous	Same	Pass
Empty Input	Error Message	Same	Pass
Invalid URL	Warning Message	Same	Pass

5.3 White Box Testing (Functional Validation)

White box testing checks internal logic, code structure, and flow of the system.

Key Areas Tested:

- Condition checking (if-else logic)
- Loop execution
- Function calls
- Database queries
- API responses

Example:

Module	Condition	Expected Result	Status
Login	Valid credentials	Login success	Pass
Login	Invalid credentials	Error message	Pass
Scan Module	Valid input	Process correctly	Pass

5.4 Test Cases

Test Case Table

Test Case ID	Description	Input	Expected Result	Actual Result	Status
TC01	User Login	correct credentials	Login successful	Same	Pass
TC02	User Login	Wrong password	Error message	Same	Pass
TC03	URL Scan	Valid URL	Risk classification	Same	Pass
TC04	Message Scan	Spam message	Detected as spam	Same	Pass
TC05	Report Threat	Submit report	Report saved	Same	Pass

5.5 Test Cases and Results

All modules were tested with different inputs and conditions.

Summary:

- All major functionalities are working correctly
- No critical bugs found
- System handles invalid inputs properly
- Outputs are accurate and consistent

Final Result:

The system has been successfully tested and is ready for deployment.

CHAPTER 6

6.1 Limitations

- Requires internet connection
 - Detection accuracy depends on dataset
 - Limited real-time threat intelligence
-

6.2 Enhancements

- AI-based threat detection
 - Browser extension integration
 - Mobile app version
 - Real-time database updates
-

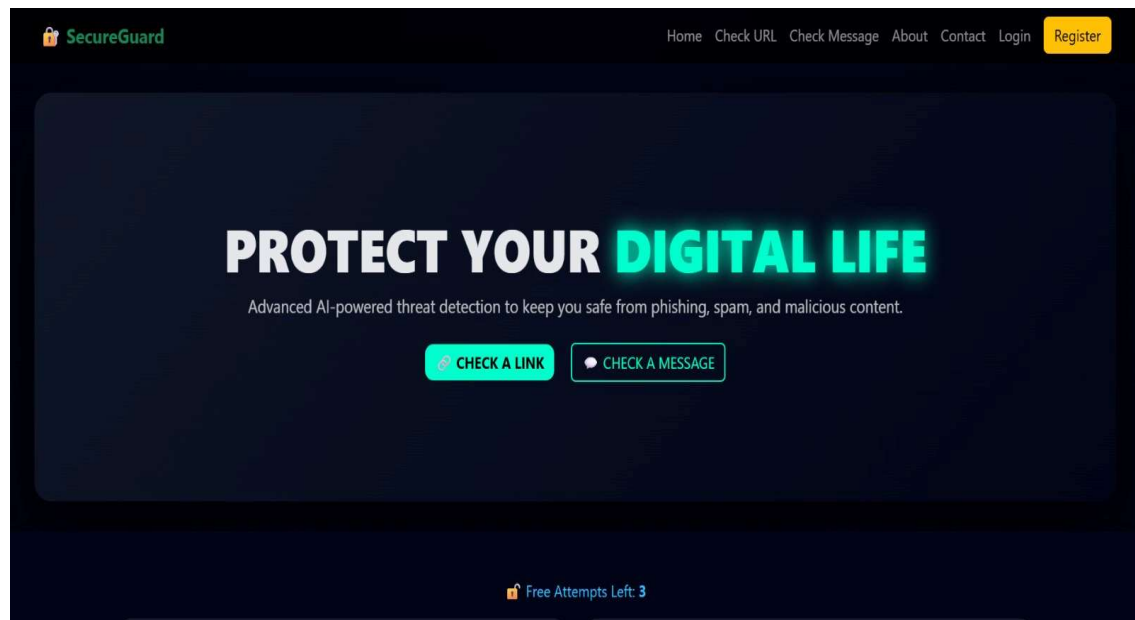
6.3 Conclusion

This project helped in understanding real-world cybersecurity challenges. The system provides an effective solution for detecting and preventing cyber threats. It improves user awareness and ensures safer online interaction.

6.4 Bibliography

- Google Research Papers
 - Microsoft Documentation (.NET)
 - SQL Server Docs
 - Cybersecurity blogs
-

Visuals :



Create Account

Join us today

Full Name

Email

Password

Confirm Password

☒ I'm not a robot

reCAPTCHA
Privacy - Terms

Register

Already have an account? [Login](#)

Welcome Back

Login to your account

Email

Password

[Forgot Password?](#)

Login

Don't have an account? [Register](#)

SecureGuard

[Home](#) [Check URL](#) [Check Message](#) [About](#) [Contact](#) [Login](#) [Register](#)

Verify OTP

Enter the code sent to your email

OTP Code

231172

Verify OTP

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URL Scan

http://malware.testing.google.test/testing/malware/

Scan Link

DANGER

Risk Assessment Complete

Risk Level: 90%

Analysis:

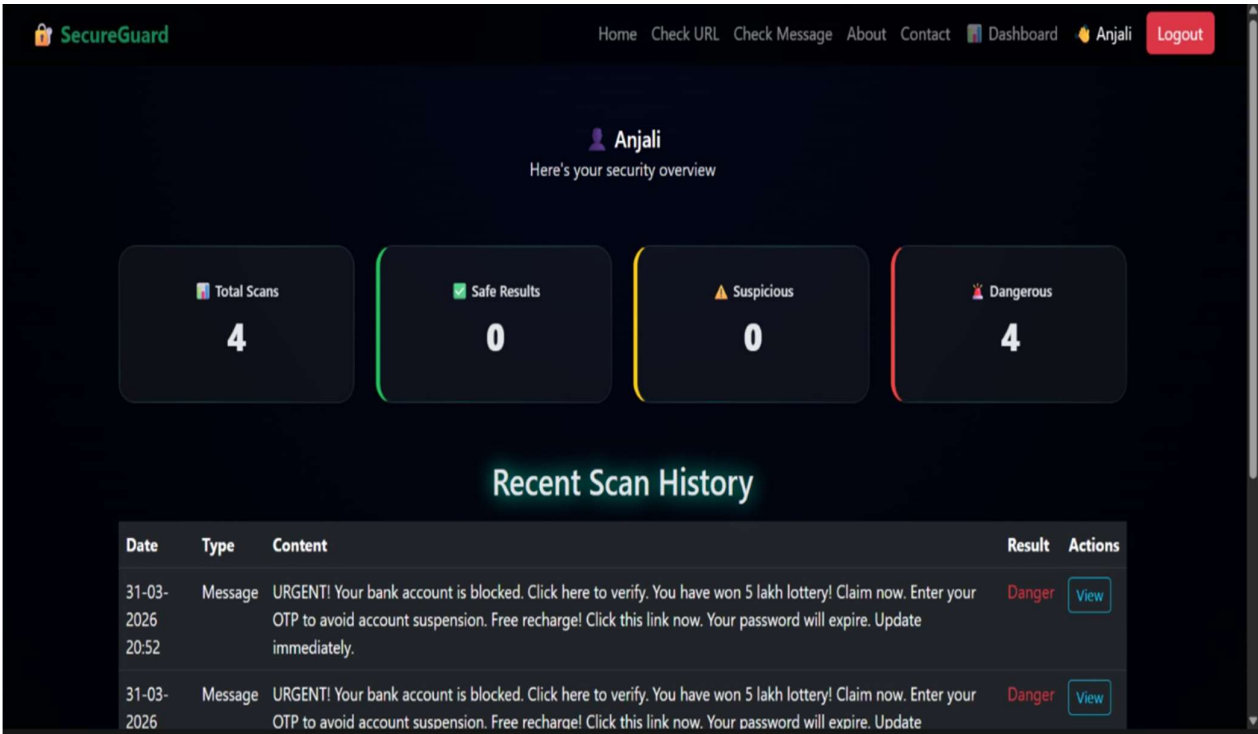
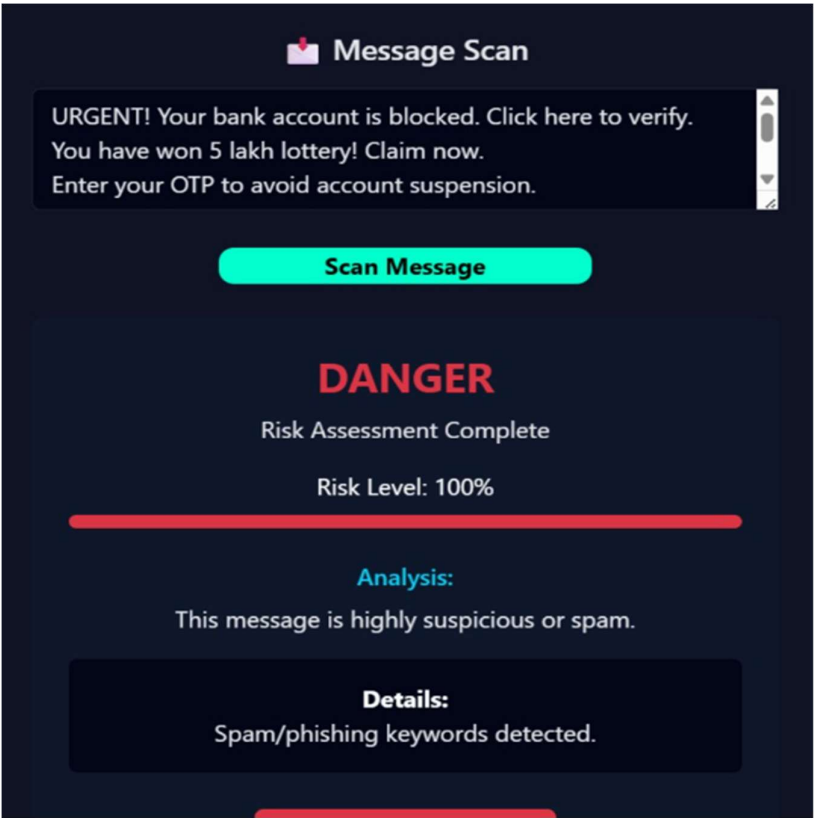
This URL is highly dangerous and may be phishing.

Details:

Malicious patterns detected. Avoid entering sensitive data.

Recommendations:

- Check domain name carefully
- Look for HTTPS (secure connection)



Recent Scans

Time	User	Type	Content	Result
31-03-2026 20:52:16	anjalijawale30@gmail.com	Message	URGENT! Your bank account is blocked. Click here to verify. You have won 5 lakh lottery! Claim now. Enter your OTP to avoid account suspension. Free recharge! Click this link now. Your password will expire. Update immediately.	Danger
31-03-2026 20:52:16	anjalijawale30@gmail.com	Message	URGENT! Your bank account is blocked. Click here to verify. You have won 5 lakh lottery! Claim now. Enter your OTP to avoid account suspension. Free recharge! Click this link now. Your password will expire. Update immediately.	Danger
31-03-2026 20:51:07	anjalijawale30@gmail.com	URL	http://malware.testing.google.test/testing/malware/	Danger
31-03-2026 20:51:07	anjalijawale30@gmail.com	URL	http://malware.testing.google.test/testing/malware/	Danger
28-03-2026 11:53:20	vivekkale.work@gmail.com	Message	URGENT! Your bank account is blocked. Click here to verify. You have won 5 lakh lottery! Claim now. Enter your OTP to avoid account suspension. Free recharge! Click this link now. Your password will expire. Update immediately.	Danger

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SecureGuard

HomeCheck URLCheck MessageAboutContactAdmin PanelAdminLogout

Admin Control Panel

System overview and threat management

Total Users

11

Total Scans

48

Reported Threats

6

Resolved

5

Reported Threats

ID	Date	Type	Content	Reporter	Status	Action
#6	28-03-2026 11:20:34	URL	http://testsafebrowsing.appspot.com/s/phishing.html	sdcloud43@gmail.com	Resolved	Blocked
#5	28-03-2026 09:15:55	URL	http://testsafebrowsing.appspot.com/s/malware.html	tmaxgamer76@gmail.com	Resolved	Blocked
#4	28-03-2026 08:51:02	URL	http://testsafebrowsing.appspot.com/s/malware.html	tmaxgamer76@gmail.com	Pending	SAFEBLOCK
#3	28-03-2026 08:49:03	URL	http://testsafebrowsing.appspot.com/s/phishing.html	tmaxgamer76@gmail.com	Resolved	Safe
#2	28-03-2026 08:19:47	URL	http://malware.testing.google.test/testing/malware/	tmaxgamer76@gmail.com	Resolved	Blocked